CS 374A Final Review

ACM @ UIUC

May 11, 2026

Disclaimers and Logistics

- **Disclaimer:** Some of us are CAs, but we have not seen the exam. We have no idea what the questions are. However, we've taken the course and reviewed previous exams, so we have **suspicions** as to what the questions will be like.
- This review session is being recorded. Recordings and slides will be distributed on EdStem after the end.
- **Agenda:** We'll quickly review all topics likely to be covered, then go through a practice exam, then review individual topics by request.
  - Questions are designed to be written in the same style as Kani's previous exams but to be *slightly* harder, so don't worry if you don't get everything right away!
- Please let us know if we're going too fast/slow, not speaking loud enough/speaking too loud, etc.
- If you have a question anytime during the review session, please ask! Someone else almost surely has a similar question.
- We'll provide a feedback form at the end of the session.

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Induction

Template

Let x be an *arbitrary* <OBJECT>. Assume for all k s.t. k is smaller than x (by <ORDERING PROPERTY>), that $P(k)$ holds.

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Some tips:

- **Always use strong induction**. All weak inductive proofs can be re-written to use strong induction with minimal changes, and the extra assumption can make your life significantly easier.
- **Write out your IH, base case, and inductive step out explicitly.** Doing so will help you avoid getting confused, and will help you avoid losing points.
- If you're performing induction on a recursive definition (strings, CFLs, etc.), generally, your inductive step will consist of one step of the recursion, and then will use IH.

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- If trying to guess whether or not a language is regular, think about memory. When processing a string through a DFA, you only need to know which state you're currently in, and do not need to look forwards/backwards in the string.
  - Implementing a DFA/NFA in code only requires $O(1)$ memory
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- If trying to guess whether or not a language is regular, think about memory. When processing a string through a DFA, you only need to know which state you're currently in, and do not need to look forwards/backwards in the string.
  - Implementing a DFA/NFA in code only requires $O(1)$ memory
  - If your checker program needs to count something without bound, the language you're checking isn't regular.
- **Regex Design Tips:** If you don't know where to start, try giving examples for strings that are in the language and strings that aren't. Look for patterns and try to build components around those patterns, then combine into something that represents the full language. Make sure to test and modify for edge cases. Explain, in English, each part of your regular expression with a short sentence. Does the explanation match the language?

DFAs/NFAs

- DFAs defined by *state set* Q, *accepting set* $A \subseteq Q$, *input alphabet* Σ, *start state* $s \in Q$, and *transition function* $\delta : Q \times \Sigma \to Q$
- NFAs allow for "trying" multiple transitions at the same time or transitioning without reading in (ϵ-transitions), accepts if there is a path to an accepting state. Transition function thereby changes to $\delta : Q \times (\Sigma \cup \{\epsilon\}) \to 2^Q$
  - Power-set construction to convert from NFA to DFA- in theory exponential-time but used in practice.
- **Tips for creating DFA/NFAs**: Break down your language into smaller patterns, and figure out what you need to store as state for each part. Make sure you clearly define all components. A drawing or transition table is just as valid as a $(Q, A, \Sigma, s, \delta)$ definition.

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Product Constructions

Given some languages $L_1, \ldots, L_n$ we want a DFA that accepts strings w satisfying $f(w \in L_1, \ldots, w \in L_n)$ where f is some logical function. Create a DFA/NFA for L using the following *rough* format:

- $Q = Q_1 \times \cdots \times Q_n$
- $\delta'(q_1, \ldots, q_n) = (\delta_1(q_1), \ldots, \delta_2(q_2))$
- $s = (s_1, \ldots, s_n)$
- $A' = \{\text{convert } f \text{ into a set expression}\}$

Fooling Sets

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- **If you see divisibility, think primes!** All primes are coprime, so primality provides for an infinite set with easier construction of distinguishing suffixes.
- If you're using strings of the form 1^k, 0^p, etc. when sampling elements of your fooling set a^i, a^j, it's completely fine to assume WLOG that $j > i$, but nothing about the underlying structure of i and j. If you want to put in such a restriction, you should instead restrict your fooling set further.

Language Transforms

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Make sure you're going in the right direction!

If you see the format $f(L) = \{k(w) : w \in L\}$, your modified NFA should be trying to *undo* k, while if you see the format $f(L) = \{w : k(w) \in L\}$, your modified NFA should be trying to *apply* k. Mixing these up is the most common mistake we see on homeworks/exams.

In some cases, only one direction is possible. For example, $\texttt{un-palin}(L) : \{w : ww^R \in L\}$ has a transformation construction, but $\texttt{palin}(L) = \{ww^R : w \in L\}$ is irregular for some L.

Context-Free Languages/Grammars

- Formally, a context-free grammar is defined by *nonterminals/variables* V, *terminals/symbols* T, *productions* P, and the *start symbol* S. Each production rule in P looks like $A \to \alpha$, where $A \in V$ and $\alpha \in (V \cup T)^*$.

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- For example, consider $V = \{S\}, T = \{0, 1\}, P = \{S \to \epsilon, S \to 0S1\}$. (You can abbreviate this to $P = \{S \to \epsilon \mid 0S1\}$.) What language is this?

Intuition

CFGs "build" strings, going from the outside in; you can choose rules to add characters on the left/right.
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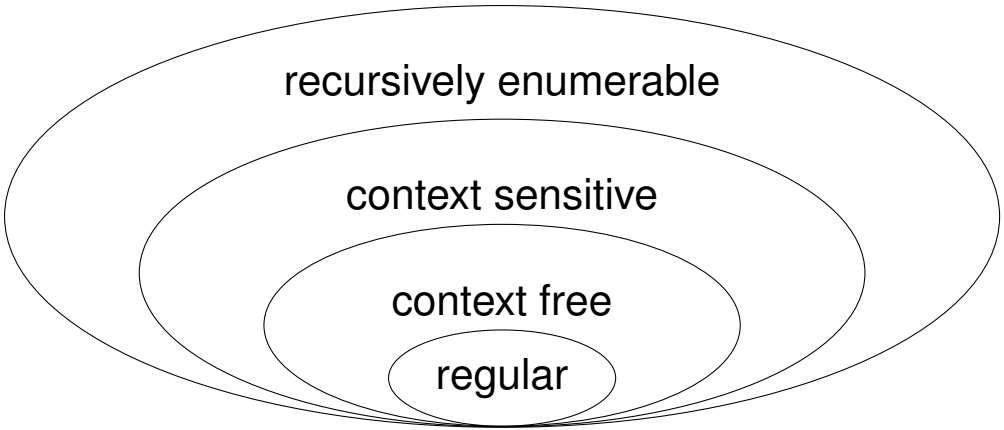

Practice Questions I

1. Let L be a regular language. Then, $L \cap \{0^n 1^n : n > 0\}$ is...
   (a) always regular
   (b) always irregular, but always context-free
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2. Define $\textsc{coverEvens}(w_1 w_2 \cdots w_n) = k_1 k_2 \cdots k_n$ and $\textsc{coverExponential}(w_1 w_2 \cdots w_n) = c_1 c_2 \cdots c_n$, where

$$k_i = \begin{cases} 1 & \text{if } i \equiv 0 \pmod 2 \\ w_i & \text{otherwise} \end{cases} \text{ and } c_i = \begin{cases} 1 & \text{if } \exists n \in \mathbf{Z} \text{ s.t. } i = 2^n \\ w_i & \text{otherwise} \end{cases}.$$

   Which of the following is true?
   (a) If L is a regular language, then $\textsc{coverEvens}(L)$ is regular.
   (b) If $\textsc{coverEvens}(L)$ is a regular language, then L is regular.
   (c) If L is a regular language, then $\textsc{coverExponential}(L)$ is regular.
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   (e) Exactly two of the above are true
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   (e) Exactly two of the above are true
   (f) Exactly four of the above are true
3. If we instead define $\textsc{UnCoverEvens}(L)$ to be $\{w : \textsc{CoverEvens}(w) \in L\}$, then would $\textsc{UnCoverEvens}(L)$ be regular for all regular L?
   (a) Yes
   (b) No

Practice Questions II

4. If L is decided by a DFA with n states, then consider the language L' consisting of all strings in L with at most 374 characters removed. Which of the following is true?
   (a) L' can be decided by a DFA with $O(n)$ states
   (b) L' can be decided by a DFA whose number of states is polynomial in n
   (c) L' can be decided by a DFA whose number of states is exponential in n
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5. Consider the language L consisting of the binary representation of all numbers congruent to 173 mod 374.
   (a) L does not have a fooling set.
   (b) L has a fooling set of size 173.
   (c) L has a fooling set of size 374.
   (d) L has a fooling set of size 375.
   (e) L has an infinite fooling set.

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   (e) L has an infinite fooling set.
6. Given a DFA M with n states, the minimum length of a string that M must accept (if $L(M)$ is non-empty) is at most:
   (a) n
   (b) $n - 1$
   (c) 2^n
   (d) $2^n - 1$
   (e) n^2

Practice Questions III

7. If L_1 is regular and L_2 is undecidable, then $L_1 \cap L_2$ is:
   (a) Always regular
   (b) Always context free
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Practice Questions III

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   (a) L^* is context-free
   (b) $L \cup \{\epsilon\}$ is context-free
   (c) $L \cap R$ is context-free for any regular language R
   (d) L^R (reverse of L) is context-free
   (e) $\{w\#w \mid w \in L\}$ is context-free

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9. Given a regular expression of length n, the equivalent minimum DFA might have
   9.1 $O(n)$ states
   9.2 $O(n^2)$ states
   9.3 $O(2^n)$ states
   9.4 $O(n!)$ states
   9.5 Always exactly n states

Practice Questions IV

1. Find a regexes for the following languages:
   (a) $\{\theta^a b^b \theta^c \mid a \geq 0 \text{ and } b \geq 0 \text{ and } c \geq 0 \text{ and } a \equiv b + c \pmod 2\}$

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2. Formally define DFAs/NFAs that accepts the following languages:
   (a) All strings whose ninth-to-last symbol is `0`, or equivalently, the set

$$\{x \circ z \mid x \in \Sigma^* \text{ and } z \in \Sigma^8\}.$$

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$$\{x \circ z \mid x \in \Sigma^* \text{ and } z \in \Sigma^8\}.$$

   (b) All strings w such that

$$(\#(\texttt{0}, w) \bmod 3) + (\#(\texttt{1}, w) \bmod 7) = (|w| \bmod 4).$$

Recursion

- **Definition:** Reducing the problem to a smaller instance of itself, where eventually we can terminate in a base case.
  - Think: If we have a problem of size n, we want to continuously reduce to a problem smaller than n.
  - Example: Tower of Hanoi

Template

```
1: procedure AmazingRecursiveAlgo(n)
2:     if n == [some base case] then
3:         return [value]
4:     else
5:         return AmazingRecursiveAlgo(n − 1)
```

- Similar to **induction**!

Recursion: Runtime Analysis

- **General Form:**

$$T(n) = \underbrace{r}_{\text{\# of subproblems}} \cdot \overbrace{T\left(\frac{n}{c}\right)}^{\text{work at each subproblem}} + \underbrace{f(n)}_{\text{work at current level}}$$

  - Describes how the amount of work changes between each level of recursion.
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\text{Decreasing: } r \cdot f(n/c) = \kappa \cdot f(n) \text{ where } \kappa < 1 &\implies T(n) = O(f(n)) \\
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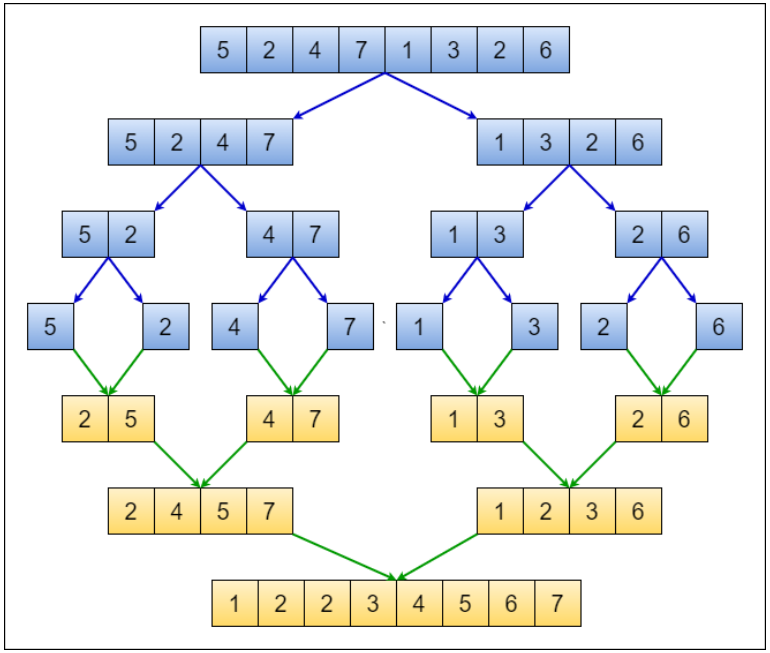
  - **Intuition:** If each level contains more work than the level below it, then the root level will dominate. If each level contains the same amount of work, then we have $log_c n$ levels with $f(n)$ work. If each level contains less work than the work below it, then the leaf nodes will dominate.

Divide and Conquer Algos: Merge Sort

- **Purpose:** Sort an arbitrary array.
- **Time Complexity**: $O(n \log n)$
- **Intuition:** Three phases: (a) split the array in half, (b) sort each side, (c) merge the sorted halves by repeatedly comparing smallest elements on each side not yet inserted.

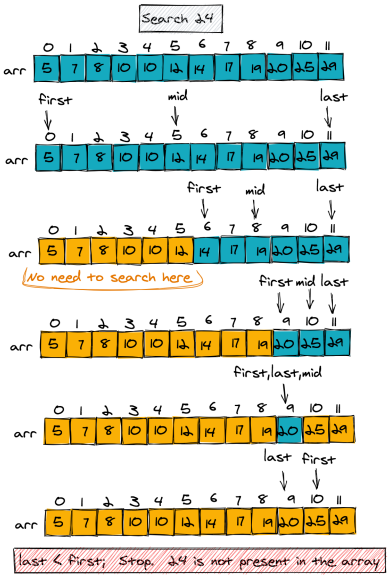


Divide and Conquer Algos: Quickselect

- **Purpose:** Get the n^{th} smallest element in an arbitrary array.
- **Time Complexity**: Avg: $O(n)$ | Worst; $O(n^2)$, ($O(n)$ with MoM)
- **Intuition**: Pick a pivot P with a value P_V and rearrange the array such that all the elements that are less than P_V are to the left of P and all the elements that are greater than P_V are to the right of P, just like quick select. If the length of the array of elements that are less than P_V is greater than n, then we know that the n^{th} smallest element is to the left of P and we recurse on the left subarray. Otherwise, we know that the n^{th} smallest element is to the right of P and we recurse on the right subarray.
  - **Why the poor worst case performance?**
  - Again, because we can get unlucky and pick the worst possible pivot at every step.
  - We can guarantee linear performance with a better pivot-picking algorithm such as MEDIANOFMEDIANS
    - Finds element that larger than $\frac{3}{10}$ and smaller than $\frac{7}{10}$ of the array's elements.
    - Runs in $O(n)$ time

Divide and Conquer Algos: Binary Search

- **Purpose:** Find the existence of an element in a sorted array
- **Time Complexity**: $O(\log n)$
- **Intuition**: Say we are trying to find the value n. Pick the middle element M in the array. If $n > M$, the element must be to the right of n and we recurse on the right. Otherwise, we recurse on the left.



Backtracking

- Technique to methodically explore the solutions to a problem via the reduction to said problem to a smaller variant of itself, a.k.a **recursion**.
- Intuitively, think of the problem space as a maze that we are trying to find the exit of. For each path, you would traverse until you reach a dead end, at which point you **back track** to try a different path.
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$$\text{LIS}(i,j) = \begin{cases} 0 & \text{if } i = 0 \\ \text{LIS}(i-1,j) & \text{if } A[i] \geq A[j] \\ \max \begin{cases} \text{LIS}(i-1,j) \\ 1 + \text{LIS}(i-1,i) \end{cases} & \text{else} \end{cases}$$

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This kind of sucks; we're redoing computation that we've already done! What if instead, we computed all the subproblems beforehand, wrote down the solutions, then did the recursion?

Dynamic Programming

- It's backtracking, but we compute all of the subproblems iteratively.
  - This idea of "writing things down" as to not repeat computation is called **memoization**

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  1. English Description
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- For a DP solution, we need:
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  6. Runtime
- **How to solve a DP:**
  - Identify how we can take advantage of a recursive call on a smaller subset of the input space.
  - Identity base cases
  - Identity recurrences (they should cover all possible cases at each step)

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Let's look at the LIS example from before: "What is the length of a longest increasing subsequence in an arbitrary array?"

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```
procedure LIS-ITERATIVE(A[1..n]):
    A ← [1 . . . n][1 . . . n]
    for all i ← 1 . . . n do
        for all j ← i . . . n do
            if A[i] ≤ A[j] then
                LIS[i][j] = 1
            else
                LIS[i][j] = 0
    for all i ← 1 . . . n do
        for all j ← 2 . . . n do
            if A[i] ≥ A[j] then
                LIS[i][j] = LIS[i − 1, j]
            else
                LIS[i][j] = max { LIS[i − 1, j]
                                { LIS[i − 1, i] + 1
    return LIS[n, n]
```

Graphs

- **Definition:** A set of vertices V connected by a set of edges E. Individual edges are notated as (u, v), where $u, v \in V$.
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- **BFS:**
  - **Purpose:** Reachability, Shortest Path (unweighted graph)
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- **Dijkstra's**
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  - **Implementation:** Dynamic Programming recurrence:
    - $d(v, k)$ is the shortest-walk distance from s to v using at most k edges
    - $d(v, k) = \min\left(d(v, k-1), \min_{u \to v} d(u, k-1) + \ell(u \to v)\right)$
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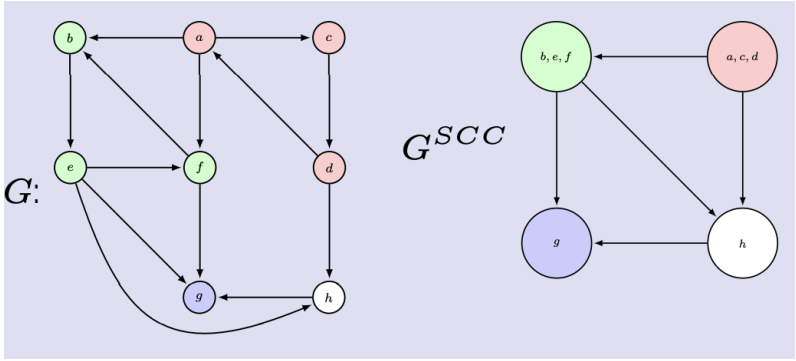
  - **Runtime:** $O(mn)$
- **Floyd-Warshall**
  - **Purpose:** APSP, yes negative edge weights
  - **Implementation:** Dynamic Programming recurrence:
    - $d(u, v, i)$ is the shortest-path distance from u to v only going through vertices $1 \ldots i$.
    - $d(u, v, i) = \min\left(d(u, v, i), d(u, i, i-1) + d(i, v, i-1)\right)$
  - **Runtime:** $O(n^3)$

Graph Algorithms: SCC

SCC-Finding Algorithms (Tarjan's, Kosaraju's)

- **Purpose:** To identify (and collapse) SCCs in a (directed) graph
- **Runtime:** $O(V + E)$
- **Returns:** A metagraph that has one node for each SCC.



Graph Algorithms: Longest Path

Longest path in a Directed Acyclic Graph (DAG)

- **Purpose:** To find the longest simple path (no repeating vertices) by weight in a graph which is guaranteed to be a DAG[1].
- **Runtime[2]:** $O(V + E)$
- **Returns:** The sum of the weights of the longest path in the DAG.

[1] Finding the longest path in other types of graphs is at least NP-hard.
[2] This is a relatively straight-forward DP on a DAG problem if you wish to derive it.

Graph Problems: General Stuff

How to solve graph problems:

1. Identify type of problem (Reachability, Shortest Path, SCC)
2. Construct new graph
   - Add sources/sinks
   - Add vertices via $V' = V \times \{\text{some set}\}$ (Useful for tracking states)
   - Add vertices via $E' = E \times \{\text{some set}\}$ (Useful for allowing/prohibit certain behaviour)
3. Apply some stock algorithm **(DO NOT MODIFY THE ALGORITHMS - MODIFY THE INPUTS!)**
4. Draw connection between how to result of the algorithm upon the new graph relates to the solution of the original question.

Graph Path Algos: When and Where

Problem Type	Weights	DAG?	Algorithm	Runtime
Shortest Path	Unweighted	DAG/No DAG	BFS	$O(n+m)$
Shortest Path	Any Weights	DAG	DP	$O(n+m)$
Shortest Path	Non-Negative Weights	No DAG	Dijkstra	$O((n+m)\log n)$
Shortest Path	Any Weights	No DAG	Bellman Ford	$O(mn)$
Longest Path	Any Weights	DAG	DP	$O(n+m)$
Longest Path	Unweighted/Any Weights	No DAG	N/A	NP Hard
APSP	Unweighted/Any Weights	No DAG/DAG	Floyd Warshall	$O(n^3)$

Where $G = (V, E)$ such that $|V| = n, |E| = m$.

Greedy Algorithms

- There's a set of standard greedy heuristics (take top scoring element, do largest/smallest task first, take element if it currently fits, etc.)
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- These heuristics make it s.t. you only consider a **subset** of the possible solutions
- To prove that a greedy optimization works, you have to prove that the subset considered **always** contains **an** optimal solution
- We do this via an exchange proof: showing that if an optimal solution exists, an equivalent solution in the checked set exists
  - This is often easy to do by designing an algorithm which transforms optimal solutions into those with the required properties, and proving that said algorithm is correct. Since we're not actually *running* the algorithm, we don't care about its runtime.

Graphs?

The adjacency matrix A of a graph G has some nice properties that are useful in applications. One such property is that $A^\ell[i, j]$ is the number of walks using ℓ edges between vertices i and j. Describe and analyze an efficient algorithm to compute A^ℓ for a graph G with n vertices. You may use the fact that two $n \times n$ matrices can be multiplied in $O(n^3)$ time.

Dancing With the Stars

Sarah loves *Dancing With the Stars*. She is very particular about the performances she highlights in the finale montage: she wants a sequence of routines that maximizes excitement, but if she pays attention to one routine, she won't be able to pay attention to the next two. Can we pick k routines for her to watch that maximizes her excitement?

There are n weekly routines, indexed 1 through n. Each routine i has an excitement score $E[i]$.
Sarah can choose to watch a routine (add $E[i]$ to the total excitement) or not pay attention to it. If she chooses to watch a routine, she can't watch the next two routines. She can choose at most k routines.

Compute a subset of of $\leq k$ routines to watch that **maximizes total excitement**.

Secret Lives of Mormon Wives

Navid's favorite show is *The Secret Lives of Mormon Wives*, which he claims to watch "for the plot" but really watches for the unhinged drama. The show revolves around MomTok, questionable business ventures, and an alarming number of MLM schemes. As a result, there are many explosive secrets (affairs, business betrayals, Instagram unfollowings), and some secrets are eventually revealed to the audience through tearful confessionals and passive-aggressive Instagram stories. The timing of these revelations affects how much total drama the audience experiences. You are in charge of choosing which secrets to reveal.

Each secret is assigned to a unique episode, and a secret can only be revealed in its assigned episode.

There are n secrets, indexed 1 through n in episode order. Each secret i has a drama value $V[i]$ if it is revealed.

If two revealed secrets $i < j$ are revealed *consecutively* among the chosen secrets and occur $k = j - i$ episodes apart, they produce a bonus drama of $k - 1$ since this increases suspense.

Suppose $n = 6$ with drama values $V = [20,\ 1,\ 30,\ 2,\ 5,\ 14]$, and we choose to reveal secrets $\{1,\ 3,\ 6\}$. Total drama:
$20 + (30 + (3 - 1 - 1)) + (14 + (6 - 3 - 1)) = 67$.

Design and analyze an algorithm to compute a subset of secrets to reveal that **maximizes the total drama**.

Bad Apple

Jeff is falling into a Bad Apple rabbit hole on Youtube. He wants to get from the original **Bad Apple** video to the video **bad apple but it's a sorting algorithm**[3] by *only* watching videos that are recommended to him in sequence.

Formally, you are given a list of n videos V such that each $v \in V$ has a watch time $\tau(v)$ and list $R(v) \subseteq V$ of sidebar recommended videos that can be watched after v is watched. Suppose video $s \in V$ is the original **Bad Apple** video and $t \in V$ is the **bad apple but it's a sorting algorithm** video. We call a sequence of videos starting from s and ending at t a *Terpsichorean marathon*.

(a) Describe and analyze an algorithm to find a *Terpsichorean marathon* of minimum total watch time.

(b) We call a video a *banger* if it is in some *Terpsichorean marathon* with minimum watch time. Design and analyze an efficient algorithm to find all the *banger* videos.

[3] Yes, this is a real thing.

Who's the number one spice?

Young Cardamom is bringing the flavor to the fish![4] His dish needs to have both **HEAT** and **SWEET** spices to be complete.

The kitchen can be modeled as a DAG where:

- Vertices represent prep stations
- Each station is labeled either HEAT, SWEET, or NEUTRAL
- A directed edge (u, v) with weight $f(u, v)$ means station u can send ingredients to station v, contributing flavor value $f(u, v)$
- Young Cardamom starts at station s and must reach plating station t

To achieve the #1 spice, Young Cardamom's path must visit **at least one HEAT station** and **at least one SWEET station**.

Design and analyze an algorithm to find a path from s to t that visits both required station types and maximizes the total flavor value.

[4] Of course, HAB is bringing the flavor to the rice.

TEAM UPDATE:

The 374 exam is being coordinated in a Signal group chat, and Pranay accidentally adds a student to the group chat. This leaks the exam to the student! Oops. You want to find out how quickly the exam can reach you.

We model the students' friend network as an unweighted, undirected[5] graph $G = (V, E)$, where each vertex is a student in the class and an edge $uv \in E$ means u and v are friends. Some students $D \subseteq V$ are in an illicit Discord server and can send the exam to each other even if there are no edges between them. However, Discord is quite a slow platform, so sending the message between two vertices in D takes $d > 1$ time. Otherwise, sending the exam between friends takes time 1. Design and analyze an algorithm to determine the shortest amount of time needed for the exam to travel from the student added to the Signal group, s, to you, t.

[5] Unlike real life, friendships in this world are always bi-directional.

how ba-a-a-ad can i be

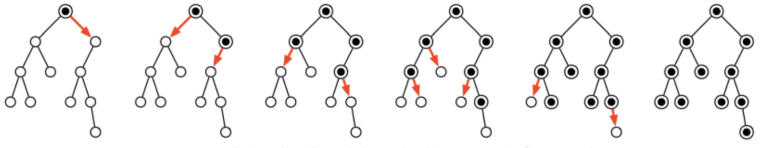
We are attempting to write a text formatter. Ideally, we want our formatter to format our text reasonably; we wouldn't want our formatter to just put each word on a new line, for example. Fortunately, we have a function badness(n), where given n characters of whitespace, it will return to us a integer of "badness". Given that each line can hold at most M character s and an array $L[1 \cdots n]$, where $L[i]$ is the length of the ith word, describe an algorithm that finds the smallest possible total badness of a given sequence of words.

the skyway sucks

You're at your office in downtown Chicago, and you're trying to get to your friend's apartment elsewhere in the city. You're given the map of Chicago as a directed graph G, with streets as edges and intersections as vertices. Each edge is weighted with the travel time (in minutes) that it takes to travel the length of the road. Some edges are annotated as moving bridges, which are only usable for certain periods (notated as intervals in minutes after you start). You can also bribe the bridge operator into moving the bridge and letting you cross by slipping him a crisp $20, in which case you can cross after 1 minute, but you only have $\$k$. Describe and analyze an algorithm to calculate the quickest route that you can go.

tell me more, tell me more, did you get very far?

Suppose we need to distribute a message to all the nodes in a given binary tree. Initially, only the root node knows the message. In a single round, each node that knows the message is allowed (but not required) to forward it to at most one of its children. Describe and analyze an algorithm to compute the minimum number of rounds required for the message to be delivered to all nodes in the tree. For example, given the following tree as input, your algorithm should return the integer 5.



A message being distributed through a binary tree in five rounds.

P and NP

- A **decision problem** is a problem with a true/false answer. (yes/no, etc.)
- **P** is the set of decision problems with a polynomial-time solver.
- **NP** is the set of decision problems with a polynomial-time *nondeterministic* solver.
- Alternatively, NP is the set of decision problems with a polynomial-time *certifier* for "true" answers, given a polynomial-size *certificate*.
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For example, consider the yes/no problem of deciding whether a graph $G = (V, E)$ has a path containing all its vertices. (Hamiltonian Path)

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Formally, an algorithm C is a certifier for problem X when $s \in X$ if and only if there exists string t such that $C(s, t) = \text{true}$.

- t here is a "certificate."
- We can show X is NP by providing this information, and showing C is polynomial-time and t is polynomial-size (with respect to the size of the input s).

co-NP

- **co-NP** is the set of decision problems X whose complements $\overline{X}$ are in NP.
- Alternatively, NP is the set of decision problems with a polynomial-time certifier for **"false"** answers, given a polynomial-size certificate.
- For example, the problem of deciding whether a graph *doesn't* have a Hamiltonian path is in co-NP.

co-NP isn't on your skillset, but be aware that this is *not* the same thing as NP.

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Make sure you're going in the right direction!

If you're trying to prove that a problem is NP-hard or undecidable, you need to reduce **from** an NP-hard/undecidable problem **to** the problem you want to prove is hard (in other words, show that an oracle for your problem can be used to solve an NP-hard/undecidable problem). The most common mistake on exams is reducing in the wrong direction.

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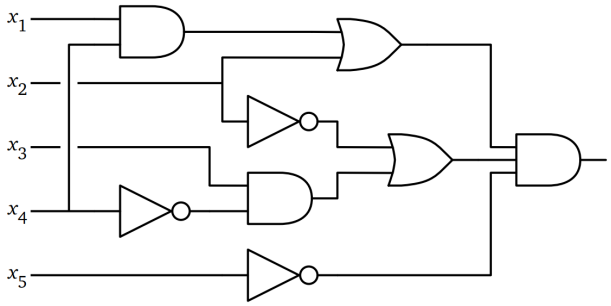
Template- Reduction

Assume that there exists an oracle function *B* which runs in [TIME CONSTRAINT]. Thus, we can solve *A* as follows:

```
1: procedure A(input):
2:     Do some preprocessing to create instances of problem B
3:     outputs ← B(generated inputs)
4:     Do some postprocessing on outputs to get the correct answer for A
```

A Tour of NP-Hard Problems: CircuitSAT and 3SAT

- **CircuitSAT**: The "original" NP-complete problem. Given a boolean circuit, is there a set of inputs that makes it return `true`?



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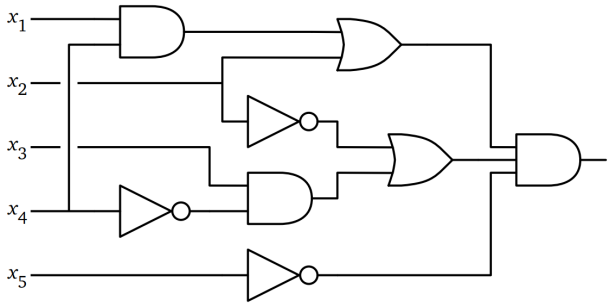

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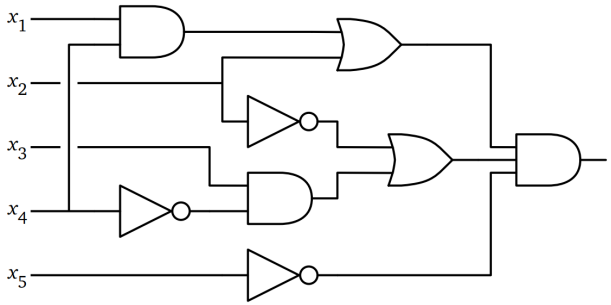

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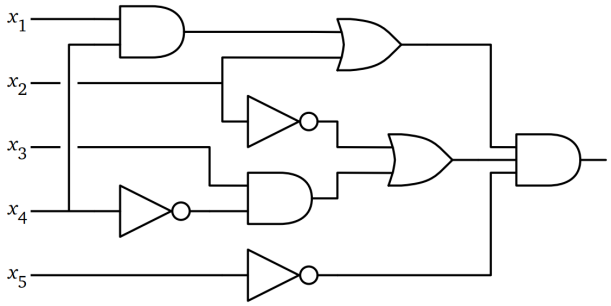

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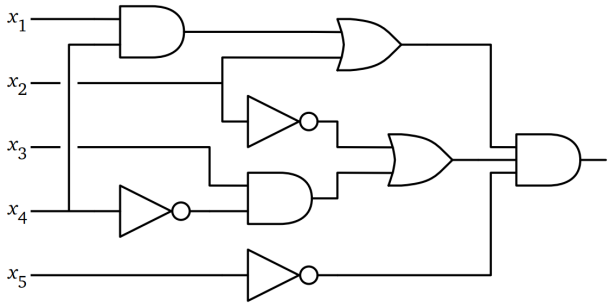

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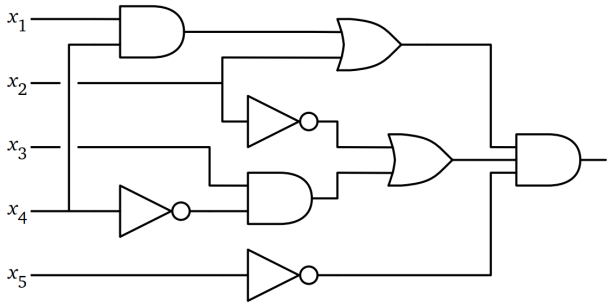

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Be careful with *k*-SAT variants!

While k-SAT for $k \geq 3$ is NP-complete, there is a polynomial-time algorithm for 2SAT. (Using strongly connected components!)

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Consider the problem **MajSAT**: Clauses now consist of 5 literals, and you must satisfy at least 3 literals in each clause. Is **MajSAT** in NP, NP-hard, both, or neither? Prove why by either stating an algorithm or providing a reduction.

A Tour of NP-Hard Problems: Max{Clique, IndSet}, MinVertexCover

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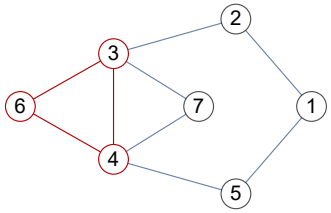

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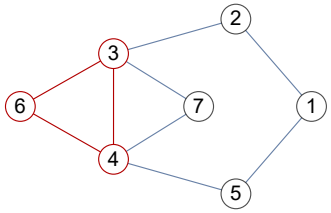

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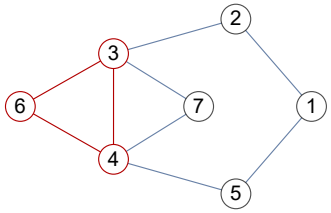

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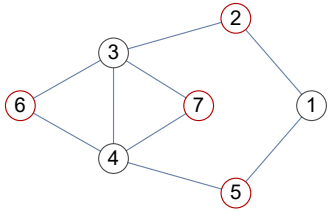

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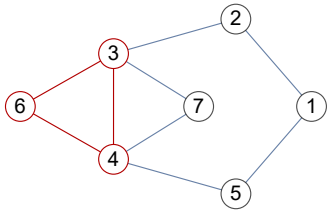

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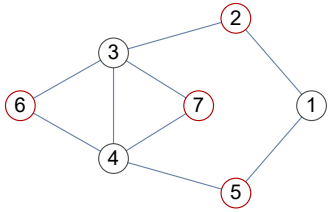

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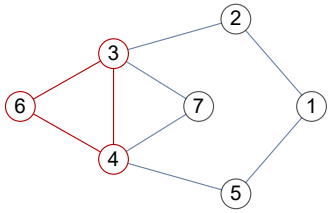

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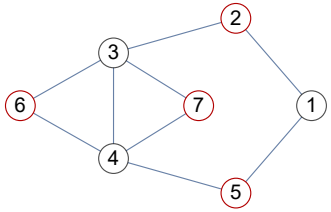

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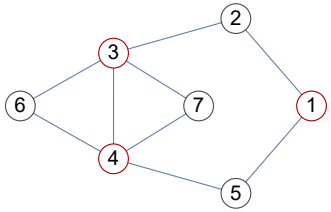
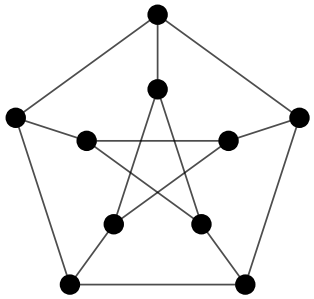


A Tour of NP-Hard Problems: Max{Clique, IndSet}, MinVertexCover

ACM is writing their review session for CS/ECE 374B MT3. While making slides, each CA writes 2 problems, either alone or in collaboration with other CAs. Since all of the CAs all have inflated egos, they won't show up to the review session unless one of the problems that they worked on is in the review session. Show that determining whether we can run a review session with at most k problems is NP-complete.

A Tour of NP-Hard Problems: Graph Coloring

- Given an (undirected) graph, can we color the nodes with at most *k* colors so that no two vertices that share an edge are of the same color?



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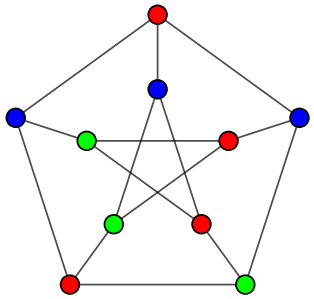

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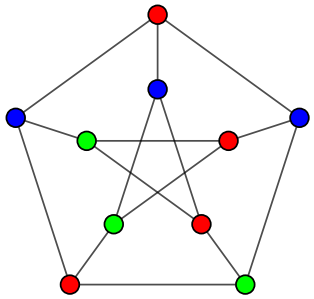

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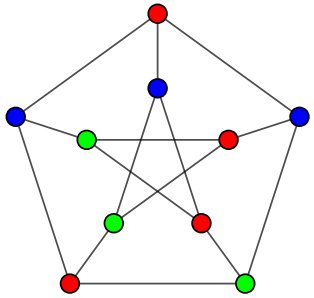

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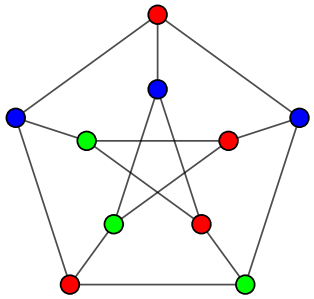

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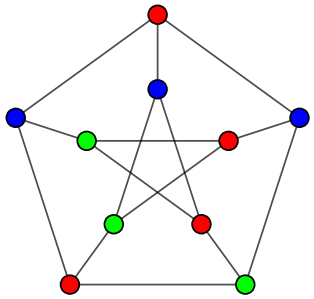

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Be careful with k-coloring variants!

While k-coloring for $k \geq 3$ is NP-complete, you can find whether a graph is bipartite (2-colorable) using DFS.

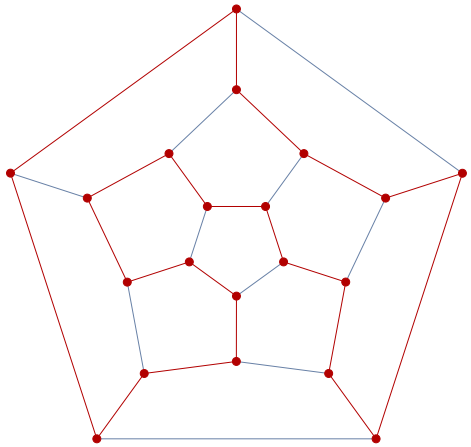
A Tour of NP-Hard Problems: Graph Coloring

A **relaxed 3-coloring** of a graph G assigns each vertex of G one of three colors (for example, red, green, and blue), such that **at most one edge** in G has both endpoints the same color.

- Give an example of a graph that has a relaxed 3-coloring, but does not have a proper 3-coloring (where every edge has endpoints of different colors).
- **Prove** that it is NP-hard to determine whether a given graph has a relaxed 3-coloring.

A Tour of NP-Hard Problems: Hamiltonian Paths and Cycles

- A **Hamilton Path** is a path that goes through each vertex *exactly* once. Likewise, a **Hamiltonian Cycle** is a cycle that goes through each node *exactly* once.
  - Every graph with a Hamiltonian cycle has a Hamiltonian path, but not every graph with a Hamiltonian path has a Hamiltonian cycle.



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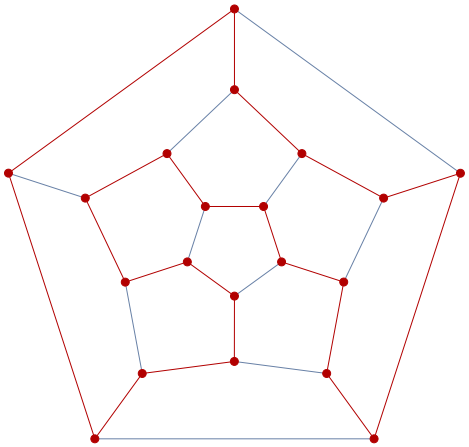

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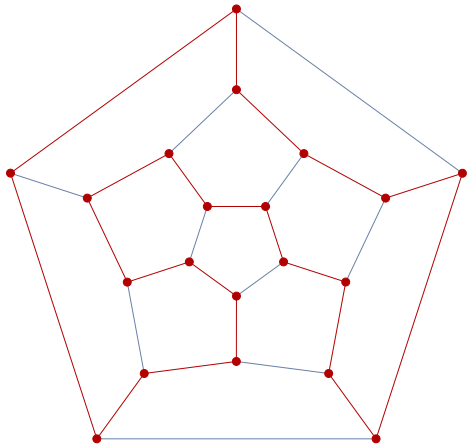

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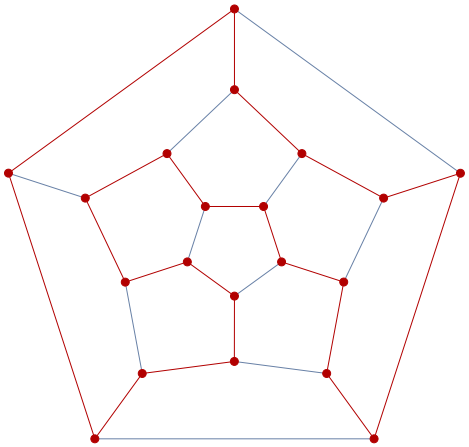

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  - You have a resource pool, and you want to use up everuthing

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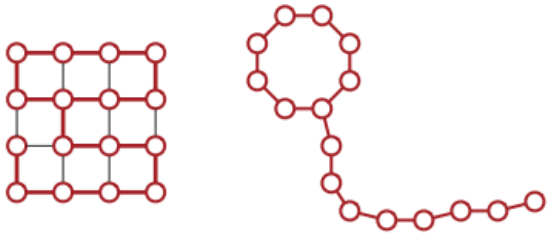
A **balloon graph** of size ℓ is a cycle of length ℓ attached to a path of length ℓ, where the cycle and the path are disjoint except for the connecting vertex. Show that it is NP-hard to determine whether a graph has a balloon subgraph of size at least k.



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- **Checkers**: given a $n \times n$ checkerboard, is there a move that captures at least k checkers?

Undecidability

- A language is **decidable** if there exists an algorithm which always returns `true` to all inputs in L and `false` to inputs not in L
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  3. Abuse the fact that you can put code into a function to derive a contradiction.

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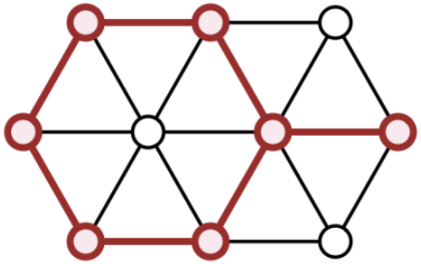
(f) NeverLeft $= \{\langle M, w \rangle : M$ never moves left on input $w\}$

twiangles are scawy

A subset S of vertices in an undirected graph G is a called triangle-free if, for every triple of vertices $u, v, w \in S$,at least one of the three edges uv, uw, vw is absent from G. Prove that find the size of teh largest triangle-free subset of vertices in a given undirected graph is NP-hard.



A triangle-free subset of 7 vertices and its induced edges.
This is ***not*** the largest triangle-free subset in this graph.

Feedback

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